

# Seth DeWhitt

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## EDUCATION

**Purdue University**, *West Lafayette, IN*

August 2023 – May 2027

*B.S. in Computer Science, Software Engineering concentration*

GPA: 3.6/4.0

**Relevant Coursework:** Data Structures, Analysis of Algorithms, Software Engineering, Systems Programming, Computer Architecture, C Programming, Software Testing, Object-Oriented Programming, Linear Algebra

**Activities:** Purdue Jazz Education Network (President), Purdue Jazz Band (Lead Tenor Saxophone)

**Awards:** Elmhurst University Jazz Festival Outstanding Soloist (2x), Dean's List (3x)

## EXPERIENCE

**Research Lead & Teaching Assistant**

August 2025 – Present

*The Data Mine, Purdue University | Git, Agile, Python, Typescript*

*West Lafayette, IN*

- Directing 11 student researchers in collaboration with the Practical AI Podcast (**1.5M listens**)
- Implementing RAG chatbot trained on **349+ episodes** with citation-backed responses
- Designed scalable podcast generation platform integrating LLM-produced transcripts and voice cloning

**Software Engineering Intern**

June 2025 – August 2025

*Chewy | Python, React, Vite, FastAPI, Docker, AWS, Terraform, Jenkins, PostgreSQL*

*Bellevue, WA*

- Deployed scalable & containerized development environments with custom tooling for in-house model execution on AWS ECS/Fargate, **reducing setup overhead by 80%** and supporting **15+ engineers**
- Automated distributed system maintenance with AWS Lambda and Kinesis, **lowering AWS expenses by 25%**
- Engineered **persistent storage architecture** linking Okta SSO authentication with AWS EFS, ECS, and RDS to support secure stateful development sessions across container restarts
- Designed and implemented Terraform-based infrastructure provisioning and Jenkins CI/CD pipelines, **decreasing deployment times by 66%**

**Undergraduate Data Science Researcher** | *Bayer Crop Sciences, Syngenta*

August 2023 – May 2025

*The Data Mine, Purdue University | Python, R, Sci-kit learn, Pandas*

*West Lafayette, IN*

- Trained a LightGBM model on **500k+ rows** to model climate change impact on corn yields at **91% accuracy**
- Optimized model performance through feature engineering and cross-validation, reducing RMSE by **24%**
- Implemented data preprocessing and QC pipelines for **100M+ data points**, validated using mixed linear models

## PROJECTS

**MyPort** | *TypeScript, OpenAI, LangChain, Supabase* | [GitHub](#) | [Link](#)

October 2025

- Pioneered RAG portfolio chatbot with Supabase, OpenAI embeddings, and LangChain retrievers
- **Reduced latency by 20%** with vector indexing and chunked doc ingestion, **supporting idempotent updates**

**Evallm** | *TypeScript, React, Next.js, Vercel, OAuth2, MongoDB* | [GitHub](#) | [Link](#)

July 2025 – August 2025

- Architected and deployed full-stack platform for **benchmarking LLM outputs**, integrating model inference with experiment tracking and analytics pipelines
- Integrated **real-time model inference**, enabling low-latency feedback and interactive evaluation workflows

**UNIX-like Shell** | *C++, Lex/Yacc*

February 2025 – April 2025

- Implemented POSIX-style shell in C++ with advanced parsing (Lex/Yacc), supporting **I/O redirection, pipelines, subshell execution, and signal handling**
- Handled process management using fork/exec & pipe syscalls, enabling **concurrent processes** and zombie reaping
- Constructed command history and environment variable expansion with **O(1) lookup**

## TECHNICAL SKILLS

**Languages:** Python, JavaScript/TypeScript, Java, C/C++, SQL, R

**Cloud/Data:** AWS, boto3, psycopg2, PostgreSQL, Pandas, Flyway

**Dev Tools:** Git, FastAPI, React, Node.js, Next.js, Vite, Tailwind, Docker, MongoDB, Terraform, Jenkins